

Reviewer Pack — q-Major Cancellation Gap Separates Permanent from Determinan...

Entry 796531c3b7fa · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-17 03:32:04 UTC

q-Major Cancellation Gap Separates Permanent from Determinant Supports

Entry ID: 796531c3b7fa **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-17 03:32:04 UTC

1. Conjecture

Field A (mathematical branch): Mahonian statistics on the symmetric group (major index, q-factorials, RSK cocharge) **Field B** (complexity object): Padded permanent/determinant comparison on shared 0/1 support matrices (GCT det-vs-perm flavor)

Statement:

For an $n \times n$ matrix M in $\{0,1\}^{\{n \times n\}}$, let $T_M = \{\sigma \in S_n : M[i, \sigma(i)] = 1 \text{ for all } i\}$ and define the q-major permanent and signed q-major determinant on the support T_M by $\text{perm}_q(M) := \sum_{\sigma \in T_M} q^{\{\text{maj}(\sigma)\}}$ and $\det_q(M) := \sum_{\sigma \in T_M} \text{sign}(\sigma) * q^{\{\text{maj}(\sigma)\}}$, where $\text{maj}(\sigma)$ is the major index (sum of descent positions). Conjecture: for every $n \geq 3$ and every 0/1 matrix M with $|T_M| \geq 2$, the $q=2$ cancellation ratio $R_2(M) := \text{perm}_2(M) / \max(1, |\det_2(M)|)$ satisfies $R_2(M) \geq \sqrt{|T_M|} / (2^n)$. Equivalently, the $\text{sign}(\sigma)$ -weighted Mahonian sum on any 0/1-support exhibits at least square-root-of-support cancellation against the unsigned version, up to a linear factor in n .

Rationale (proposer's reasoning):

Permanent vs determinant in GCT differs precisely by replacing $\text{sign}(\sigma)$ with 1 inside the S_n -sum; weighting both sums by $q^{\{\text{maj}(\sigma)\}}$ (a Schur-positive Hecke-algebra principal-specialization) refracts that difference through irreducible S_n -character spectra, and at $q=2$ one already sees $[n]_2! = \prod(2^k - 1)$ for perm_n but a much smaller signed sum for $\det_n$. A square-root cancellation lower bound on every common support T_M would certify that determinantal coefficient sequences on permanental supports always look 'random-like' relative to their unsigned partners, giving a Mahonian witness for the per-vs-det gap that is preserved under support-preserving substitutions.

Taxonomy category: NISAN_WIGDERSON_DESIGNS (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): ea7d4d6319e7c39d

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

Across n in $\{3..10\}$ with 30 seeds each (240 trials), retain only matrices with $|T_M| \geq 2$.
Conjecture is SUPPORTED iff every retained trial satisfies $R_2(M) \geq \sqrt{|T_M|}/(2n)$;
FALSIFIED iff any single retained (n, seed) yields $R_2(M) < \sqrt{|T_M|}/(2n)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.95	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.90	SAFE	SAFE
ALGEBRIZATION	SAFE	0.90	SAFE	SAFE
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): -major index permanent determinant 0/1 matrix q-analog -signed Mahonian sum descent statistic support symmetric group cancellation - q-permanent q-determinant geometric complexity theory major index

Top relevant hits considered: - [http://arxiv.org/abs/1503.00822v2] Product Ranks of the 3×3 Determinant and Permanent - [http://arxiv.org/abs/2302.11221v6] A q-analog of certain symmetric functions and one of its specializations - [http://arxiv.org/abs/1302.4419v3] The determinant of the Malliavin matrix and the determinant of the covariance matrix for multiple integrals - [http://arxiv.org/abs/0810.2731v3] Fix-Euler-Mahonian statistics on wreath products - [http://arxiv.org/abs/1101.4332v2] Mahonian Pairs - [http://arxiv.org/abs/2402.03644v2] Signed Mahonian Polynomials on Derangements in Classical Weyl Groups - [http://arxiv.org/abs/1611.00827v2] Geometric complexity theory and matrix powering - [http://arxiv.org/abs/2504.04416v1] Meta-Mathematics of Computational Complexity Theory

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90s$ wall time per cycle **Execution:** rc=1, elapsed=1.4s

5.1 Generated Python source

```
import random
import math
from itertools import permutations

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = random.randint(3, 10)
    M = [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]

    T_M_size = sum(all(M[i][sigma[i]] == 1 for i in range(n)) for sigma in permutations(range(n)))
    if T_M_size < 2:
        return {
            "metric_name": "R_2",
```



```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in <genexpr>
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j])) for sigma in permutations
```

```
NameError: name 'j' is not defined
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed with a NameError ('j' undefined) in the major-index computation before any trial completed, so no data was produced to evaluate the pre-registered criterion. | next: Fix the maj(sigma) computation to correctly count descent positions (i where sigma[i] > sigma[i+1]) and rerun the 240-trial sweep.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	claude_max	opus	0	0	323574
2	preregistration	claude_max	opus	0	0	8188
3	novelty	claude_max	opus	0	0	10064
4	novelty	claude_max	opus	0	0	9261
5	test_gen	mistral	codestral-latest	0	0	305776
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	295712
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	288124
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	180958
9	judge	claude_max	opus	0	0	3652

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 1425309 ms total latency. Provider mix: {'claude max': 5, 'mistral': 1, 'ollama remote': 3}

(full prompt+response transcripts available in [research/audit/796531c3b7fa.jsonl](#))

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/796531c3b7fa.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp sandbox/test *.py (per-cycle)

Replay tarball: [research/replay/796531c3b7fa.tar.gz](#) (if generated)